Lai Hao Ran

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Education

2015–2018 Ph.D., Ecology and Evolutionary Biology, National University of Singapore. Advisors: Michiel van Breugel & Hugh Tan Tiang Wah. Thesis: *Functional ecology of tropical secondary forests*

2009–2012 B.Sc. (Hons.) Ecology, The University of Queensland. Advisor: Margaret M. Mayfield. Thesis: *Functional recovery following logging in subtropical forests*

Employment

2024–present Data scientist, South East Asia Rainforest Research Partnership (SEARRP), Malaysia
— Virtual Ecosystem

2022–present Research Fellow, Bioprotection Aotearoa, School of Biological Sciences, University of Canterbury, New Zealand
— Transferability of trait-based species distribution models
— Evo-ecological trade-offs in crop–pest systems

2019–2021 Research Fellow, Centre for Integrative Ecology, School of Biological Sciences, University of Canterbury, New Zealand
— Species interactions and tree growth in tropical and temperate forests

2019 Visiting Research Fellow, Centre for Urban Greenery and Ecology, National Parks Board, Singapore
— Tree diameter growth in urban streetscapes

2018–2019 Research assistant, Department of Biological Sciences, National University of Singapore, Singapore
— Data analysis and freshwater swamp forest survey

2015–2018 Teaching assistant, National University of Singapore, Singapore

2015 Biology Olympiad trainer, National Junior College, Singapore

2012–2014 Research assistant, Mayfield Plant Ecology Lab, The University of Queensland, Australia
— Plant community dynamics in subtropical forests and annual grassland

2012–2014 Research officer, Centre for Mined Land Rehabilitation, The University of Queensland, Australia
— Map and identify threats of coal mining to an endangered plant species

2012–2014 Research assistant, Buckley Ecology Lab, The University of Queensland, Australia

2012–2014 Course Tutor, The University of Queensland, Australia

Teaching

2025 Bayesian statistics for ecologists

2015–2018 Field Studies in Biodiversity, Ecology of Terrestrial Environments, Tropical Horticulture, Quantitative Methods for Ecological Research, Comparative Botany, Ecology and the Environment, Biodiversity

2012–2014 Biostatistics, Ecology, Plant Biology

Supervision & Mentoring

Ongoing	Pin Jia Chan (PhD), Daniel Hernández (PhD), Rose Foster-Dyer (PhD), Felicia Wei Shan Leong (PhD), Tim Logan (MSc.)
2024	Sophie O'Brien (PhD), Gabrielle Koerich (PhD), Jack Anderson (MSc.)
2023	Laura van Ginkel (MSc.), Rory Lennox (MSc.)
2017	Zhong Yu Chiam (BSc. Hons)
2016	Melissa Wong Qi Yun (BSc. Hons), Amanda Ng Wei Ling (BSc. Hons), Pin Jia Chan (Undergraduate programme), Zhong Yu Chiam (Undergraduate programme), Rens Gerard Brouwer (Undergraduate programme), Srishti Arora (Undergraduate programme)
2015	Germaine Tan Su Yin (BSc. Hons)

Publications

Peer-reviewed articles

Lai, H. R. (2025). Model-based ordination for phenological studies: From controlling sampling bias to inferring temporal associations. *Methods in Ecology and Evolution*, 16, 1542–1546.

Chan, P. J., Lai, H. R., Burns, B. R., Schwendenmann, L. (2025). Potential effects of an emerging disease on tree communities around a susceptible foundation tree species. *Austral Ecology*, 50, e70102.

Herbert, S. M., Tomscha, S. A., Lai, H. R., Benavidez, R., Balkwill, C. G., Ruston, P. R., Jackson, B., Deslippe, J. R. (2025). Identifying potentially suitable and accessible refugia to mitigate impacts of an emerging disease on a rare tree. *Conservation Biology*, e70088. https://doi.org/10.1111/cobi.70088

Dehling, D. M., Lai, H. R., Stouffer, D. B. (2025). Eltonian niche modelling: applying joint hierarchical niche models to ecological networks. *Ecology Letters*, 28(6), e70120.

Lai, H. R., Hill, T., Stivanello, S., Chapman, H. M. (2025). Changes in quantity and timing of foliar and reproductive phenology of tropical dry-forest trees under a warming and drying climate. *Journal of Ecology*, 113(6), 1560–1572.

Lai, H. R., Chong, K. Y., Wong, M. Q. Y., Tan, H. T. W., Yeo, D. C. J. (2025). Litter Depth, Not Canopy Openness, Mediates the Occurrence of an Invasive Shrub From the Forest Edge to Interior. *Biotropica*, 57, e70002.

Lai, H. R., Bellingham, P. J., McCarthy, J. K., Richardson, S. J., Wiser, S. K., Stouffer, D. B. (2024). Detecting Nonadditive Biotic Interactions and Assessing Their Biological Relevance among Temperate Rainforest Trees. *American Naturalist*, 204(2), 105–120.

O'Brien, S. A., Anderson, D. P., Lavorel, S., Lai, H. R., de Burgh, N., Tylianakis, J. M. (2024). Landscape patterns drive provision of nature's contributions to people by mobile species. *Journal of Applied Ecology*, 61, 2666–2678.

Martins, L. P., Garcia-Callejas, D., Lai, H. R., Wootton, K. L., Tylianakis, J. M. (2024). The propagation of disturbances in ecological networks. *Trends in Ecology and Evolution*, 39(6), 558–570.

Radford Smith, J., Lai, H. R., Cathcart-van Weeren, E., Dwyer, J. (2024). A Functional Basis for the Assembly of Australian Subtropical Rainforest Tree Communities. *Ecology Letters*, 27, e70014.

Lai, H. R., Chong, K. Y., Yee, A.T.K. (2022). Ten years after: what we learned from the Mandai storm forest. *Nature in Singapore*, Supplement(e2022121), 207–218.

Hall, J. S., Plisinski, J. S., Mladinich, S. K., van Breugel, M., Lai, H. R., Asner, G. P., Walker, K., Thompson, J. R. (2022). Deforestation scenarios show the importance of secondary forest for meeting Panama's carbon goals. *Landscape Ecology*, 37(3), 673–694.

Lam, W. N., Chan, P. J., Ting, Y. Y., Sim, H. J., Lian, J. J., Chong, R., Rahman, N. E., Tan, L. W. A., Ho, Q. Y., Chiam, Z., Arora, S., Lai, H. R., Ramchunder, S. J., Peh, K. S. H., Cai, Y., Chong, K. Y. (2022). Habitat Adaptation Mediates the Influence of Leaf Traits on Canopy Productivity: Evidence from a Tropical Freshwater Swamp Forest. *Ecosystems*, 25(5), 1006–1019.

Lai, H. R., Chong, K. Y., Yee, A. T. K., Mayfield, M. M., Stouffer, D. B. (2022). Non-additive biotic interactions improve predictions of tropical tree growth and impact community size structure. *Ecology*, 103(2), e03588.

Lai, H. R., Tan, G. S. Y., Neo, L., Kee, C. Y., Yee, A. T. K., Tan, H. T. W., Chong, K. Y. (2021). Decoupled responses of native and exotic tree diversities to distance from old-growth forest and soil phosphorus in novel secondary forests. *Applied Vegetation Science*, 24(1), e12548.

Lai, H. R., Craven, D., Hall, J. S., Hui, F. K. C., van Breugel, M. (2021). Successional syndromes of saplings in tropical secondary forests emerge from environment-dependent trait–demography relationships. *Ecology Letters*, 24(9), 1776–1787.

Falster, A. D., Gallagher, R., Wenk, E., Wright, I., Indiarto, D., ..., Lai, H. R., ... Kasia Ziemińska. (2021). AusTraits – a curated plant trait database for the Australian flora. *Nature Scientific Data*, 8(1), 254.

Song, X. P., Lai, H. R., Wijedasa, L. S., Tan, P. Y., Edwards, P. J., Richards, D. R. (2020). Height-diameter allometry for the management of city trees in the tropics. *Environmental Research Letters*, 15(11), 114017.

Lai, H. R., Chong, K. Y., Yee, A. T. K., Tan, H. T. W., van Breugel, M. (2020). Functional traits that moderate tropical tree recruitment during post-windstorm secondary succession. *Journal of Ecology*, 108(4), 1322–1333.

Yee, A. T. K., Lai, H. R., Chong, K. Y., Neo, L., Koh, C. Y., Tan, S. Y., Seah, W. W., Loh, J. W., Lim, R. C. J., van Breugel, M., Tan, H. T. W. (2019). Short-term responses in a secondary tropical forest after a severe windstorm event. *Journal of Vegetation Science*, 30(4), 720–731.

Chiam, Z., Song, X. P., Lai, H. R., Tan, H. T. W. (2019). Particulate matter mitigation via plants: Understanding complex relationships with leaf traits. *Science of the Total Environment*, 688, 398–408.

van Breugel, M., Craven, D., Lai, H. R., Baillon, M., Turner, B. L., Hall, J. S. (2019). Soil nutrients and dispersal limitation shape compositional variation in secondary tropical forests across multiple scales. *Journal of Ecology*, 107(2), 566–581.

Wainwright, C. E., HilleRisLambers, J., Lai, H. R., Loy, X., Mayfield, M. M. (2019). Distinct responses of niche and fitness differences to water availability underlie variable coexistence outcomes in semi-arid annual plant communities. *Journal of Ecology*, 107(1), 293–306.

Lam, W. N., Lai, H. R., Lee, C. C., Tan, H. T. W. (2018). Evidence for pitcher trait-mediated coexistence between sympatric Nepenthes pitcher plant species across geographical scales. *Plant Ecology and Diversity*, 11(3), 283–294.

Bimler, M. D., Stouffer, D. B., Lai, H. R., Mayfield, M. M. (2018). Accurate predictions of coexistence in natural systems require the inclusion of facilitative interactions and environmental dependency. *Journal of Ecology*, 106(5), 1839–1852.

Lai, H. R., Hall, J. S., Batterman, S. A., Turner, B. L., van Breugel, M. (2018). Nitrogen fixer abundance has no effect on biomass recovery during tropical secondary forest succession. *Journal of Ecology*, 106(4), 1415–1427.

Wainwright, C. E., Staples, T. L., Charles, L. S., Flanagan, T. C., Lai, H. R., Loy, X., Reynolds, V. A., Mayfield, M. M. (2018). Links between community ecology theory and ecological restoration are on the rise. *Journal of Applied Ecology*, 55(2), 570–581.

Sams, M. A., Lai, H. R., Bonser, S. P., Vesk, P. A., Kooyman, R. M., Metcalfe, D. J., Morgan, J. W., Mayfield, M. M. (2017). Landscape context explains changes in the functional diversity of regenerating forests better than climate or species richness. *Global Ecology and Biogeography*, 26(10), 1165–1176.

Lai, H. R., Hall, J. S., Turner, B. L., van Breugel, M. (2017). Liana effects on biomass dynamics strengthen during secondary forest succession. *Ecology*, 98(4), 1062–1070.

Lai, H. R., Mayfield, M. M., Gay-des-combes, J. M., Spiegelberger, T., Dwyer, J. M. (2015). Distinct invasion strategies operating within a natural annual plant system. *Ecology Letters*, 18(4), 336–346.

Preprints

O'Brien, S. A., Tylianakis, J. M., Anderson, D. P., Boesing, A. L., Lai, H. R., Provost, G. Le, Manning, P., Neyret, M., Blüthgen, N., Jungi, K., Magdon, P., Müller, S., Scherer-Lorenzen, M., Lavorel, S. (2025). Mobile species' responses to surrounding land use generate trade-offs among nature's contributions to people. *BioRxiv*.

Lennox, R. S., McIntosh, A. R., Lai, H. R., Stouffer, D. B., Boddy, N. C., Zammit, C., Tonkin, J. D. (2024). Introduced trout hinder the recovery of native fish following an extreme flood disturbance. *BioRxiv*.

Lai, H. R., Cheeseman, A. W., Hall, J. S., Slot, M., Breugel, M. van. (2024). Species- and community-level demographic responses of saplings to drought during tropical secondary succession. *EcoEvoRxiv*.

Lai, H. R., Burcham, D. C., Wang, J. W., Stouffer, D. B., Qiu, D. W., Yee, A. T. K. (2024). Quantifying life-history trade-offs in diameter growth for tropical tree species from a large urban inventory dataset. *EcoEvoRxiv*.

Software

Lai, H. R.. (2025) `SGtraits`: A trait database for the Singapore flora (v0.0.2). *Zenodo*.

Virtual Ecosystem Project Team. (2024) `Virtual Ecosystem`.

Lai, H. R., Chong K. Y., Yee A. T. K. (2020) `novelforestSG`: Data and model for Lai, Chong et al. (2020) *Appl. Veg. Sci.* R package version 2.0.

Song, X. P., Lai, H. R. (2020) `allometree`: Allometric Scaling of Urban Trees. R package version 0.1.

Conferences & Presentations

2025	Invited talks. National University of Singapore and National Parks Board Singapore.
2024	Ecological Society of Australia. Melbourne, Australia.
2022	sUCCESS secondary forest workshop. Leipzig, Germany.
2022	Association of Tropical Biology and Conservation (ATBC). Cartagena, Colombia.
2020	International Statistical Ecology Conference (ISEC). virtual.
2018	Association of Tropical Biology and Conservation (ATBC). Kuching, Malaysia.
2016	Conservation Asia. Singapore.
2015	20th Biological Science Graduate Congress. Bangkok, Thailand.

Awards & Scholarships

2022–2025	Marsden Fast Start Fund (NZD 360,000), Royal Society Te Apārangi, New Zealand
2019	International Society for Plant Molecular Biology Medal for most outstanding thesis
2019	CUGE Visiting Research Fellowship, National Parks Board Singapore
2017 & 2018	Teaching Assistant Award, National University of Singapore
2016	City Developments Limited (CDL) Urban Ecology and Conservation Scholarship
2013	University Medal, University of Queensland
2011	Summer Research Scholarship, University of Queensland
2010	D.A. Herbert Prize in Botany & F.A. Perkins Prize in Entomology, University of Queensland

Services & Outreach

Membership	Association of Tropical Biology and Conservation (ATBC), Open Traits Network (OTN), Ecological Forecast Initiative (EFI)
Peer review	*Science*, *Journal of Ecology*, *Functional Ecology*, *Ecology*, *Methods in Ecology and Evolution*, *Ecological Applications*, *Ecography*, *Biodiversity and Conservation*, *Biological Invasions*, *Science of the Total Environment*, *Journal of Vegetation Science*, *Global Ecology and Biogeography*, *Global Change Biology* etc.
2020	`RMarkdown` tutorial, University of Canterbury, New Zealand
2017	Science Research Programme, Hwa Chong Institution, Singapore
2016	Festival of Biodiversity, Singapore Botanic Gardens

Languages

English (professional working proficiency); Mandarin (mother tongue); Cantonese (regional dialect); Malay (national language)

Other skills

`R` statistical and programming language, Bayesian inference, version control (GitHub), plant survey and identification, geographic information system (GIS)

Professional references

Robert Ewers (current supervisor)
Professor, Imperial College London, United Kingdom
South East Asia Rainforest Research Partnership (SEARRP), Malaysia
Tel: +44 020 7594 2223
Email: `r.ewers@imperial.ac.uk`

Jason Tylianakis (current supervisor)
Professor, University of Canterbury, New Zealand
Tel: +64 3369 5379
Email: `jason.tylianakis@canterbury.ac.nz`

Michiel van Bruegel (former PhD supervisor & ongoing collaborator)
Associate Professor, National University of Singapore, Singapore
Tel: +65 6601 3705
Email: `michiel.vanbreugel@yale-nus.edu.sg`

Personal reference

Nik Fadzly bin N Rosely (collaborator)
Associate Professor, Universiti Sains Malaysia, Malaysia
Tel: +60 4653 5489
Email: `nfadzly@usm.my`